

COMP3242/6242: Deep Learning

Week 9 — Contrastive Learning

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Overview

- ▶ representation learning
- ▶ contrastive learning loss functions
- ▶ momentum and exponential moving averages
- ▶ avoiding collapse
- ▶ applications

Reading and Resources.

- ▶ Weinberger and Saul, "Distance metric learning for large margin NN classification," 2006.
- ▶ Hadsell et al., "Dimensionality reduction by learning an invariant mapping," 2006.
- ▶ Radford et al., "Learning transferable visual models from nat. lang. supervision," 2021.
- ▶ Caron et al., "Emerging properties in self-supervised vision transformers," 2021.

Big Picture: Representation Learning

- ▶ supervised learning requires labels, which are expensive to collect
- ▶ many domains have massive amounts of unlabeled data
- ▶ useful to learn **embeddings** (aka latent representations) without labels and apply these to general inference tasks later
 - ▶ e.g., by fine-tuning or with a smaller downstream model
- ▶ often based on a **self-supervised** objective

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 - ▶ e.g., by fine-tuning or with a smaller downstream model
- ▶ often based on a **self-supervised** objective
 - ▶ the simplest (and earliest) is auto-encoding
 - ▶ others can be derived from various **pretext** tasks, e.g., predict image rotations, colourization, or completion (language or image)
 - ▶ predict next video frame or next word in a sentence

Contrastive Learning

- ▶ **idea:** learn representations from raw data such that
 - ▶ similar objects are close together in embedding space
 - ▶ dissimilar objects are far apart
- ▶ consider an input x with similar example x^+ and dissimilar example x^-
- ▶ mathematically, we embed inputs into a latent space $z = f(x; \theta)$ such that

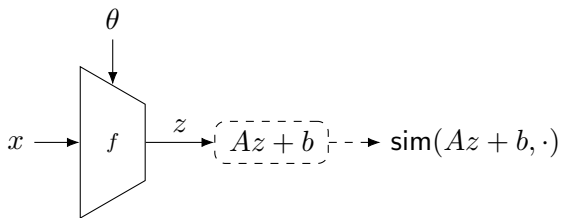
$$\text{sim}(z, z^+) \gg \text{sim}(z, z^-)$$

for some similarity measures $\text{sim}(\cdot, \cdot)$

- ▶ we typically use cosine similarity
- ▶ f is called the embedding function or encoder
- ▶ sometimes f includes a linear projection head, which is removed after training

Projection Heads

- ▶ trainable projection heads are sometimes used during training
- ▶ they can be thought of as part of the similarity metric but are not part of the learned representation



Euclidean Distance and Cosine Similarity

- ▶ the angle between two vectors $x, y \in \mathbb{R}^n$ is given by

$$\theta = \cos^{-1} \left(\frac{x^T y}{\|x\| \|y\|} \right)$$

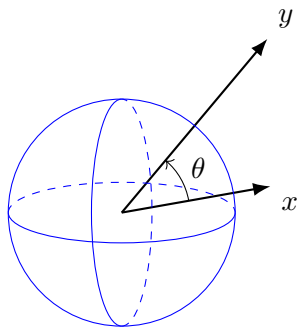
- ▶ we define **cosine similarity** as

$$\text{sim}(x, y) = \cos(\theta) = \frac{x^T y}{\|x\| \|y\|}$$

to capture how well two vectors are aligned

- ▶ if the vectors are normalized, $\|x\| = \|y\| = 1$, then cosine similarity is the same as their (negative scaled and shifted) square-distance

$$d(x, y)^2 = \|x - y\|^2 = \|x\|^2 - 2x^T y + \|y\|^2 = 2 - 2x^T y$$



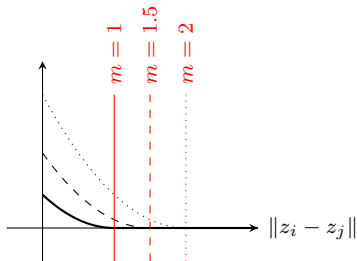
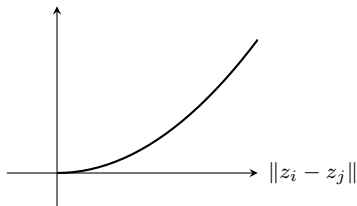
Contrastive Loss

(Hadsell et al., 2006)

- pairwise training $(x_i, x_j) \rightarrow (z_i, z_j)$,

$$\ell(\theta; z_i, z_j) = \begin{cases} \|z_i - z_j\|^2, & \text{if similar} \\ (\max\{0, m - \|z_i - z_j\|\})^2, & \text{if dissimilar} \end{cases}$$

for some margin hyperparameter m



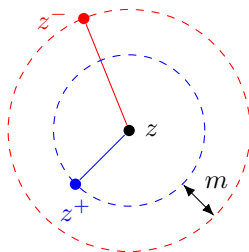
Triplet Loss

(Schroff et al., 2015; Weinberger and Saul, 2006)

- ▶ define a triplet (x, x^+, x^-) containing an **anchor** example, and a positive and negative example (wrt the anchor)
- ▶ the triplet loss is defined as

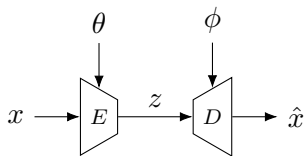
$$\ell = \max\{0, \|z - z^+\|^2 - \|z - z^-\|^2 + m\}$$

- ▶ easy triplets already have the negative far away, so $\ell = 0$
- ▶ hard triplets have the negative example closer to the anchor than the positive and incur a large penalty
- ▶ in order to learn we may need to **mine** the dataset for hard examples

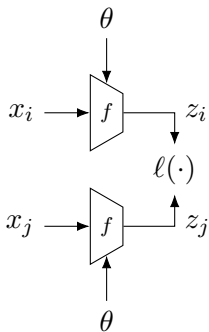


As Network Diagrams

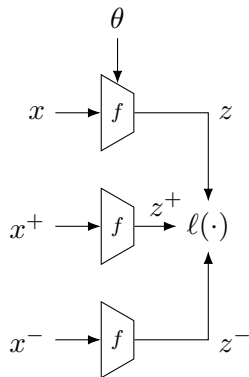
**auto-encoder
(unary loss)**



pairwise loss



triplet loss



- we typically train θ for only the anchor branch when learning with triplet loss

InfoNCE Loss

(van den Oord et al., 2018)

- ▶ **problem:** the triplet loss is data inefficient
- ▶ **solution:** use pairwise loss and assume everything else in the batch is negative

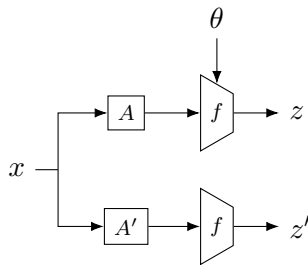
$$\ell = -\log \left(\frac{\exp \text{sim}(z_i, z_j)}{\sum_{k \neq i} \exp \text{sim}(z_i, z_k)} \right)$$

- ▶ compare with multi-class logistic model and cross-entropy loss (from Week 2)
- ▶ usually includes a temperature parameter τ
 - ▶ as $\tau \rightarrow 0$ probabilities become sharp and hard negatives dominate the loss
 - ▶ this results in large gradients making training unstable
 - ▶ so annealed during training, starting high and ending low

SimCLR

(Chen et al., 2021)

- ▶ where do similar pairs come from?
- ▶ use random augmentation
 - ▶ e.g., for images apply rotations, crops, skews, masks, etc.
- ▶ can be done on-the-fly
- ▶ for each element in the batch we generate 1 positive pair and $2(B - 1)$ negatives



Momentum Contrast (MoCo)

(He et al., 2019)

- ▶ even with InfoNCE loss, many negatives are needed, hence large batch sizes
- ▶ MoCo maintains a queue of previous embeddings z_k to use as negatives

$$\ell = -\log \left(\frac{\exp \text{sim}(z_i, z_j)}{\underbrace{\sum_{k \neq i} \exp \text{sim}(z_i, z_k)}_{\text{from mini-batch}} + \underbrace{\sum_{z \in Q} \exp \text{sim}(z_i, z)}_{\text{from queue}}} \right)$$

- ▶ however old embeddings in the queue have drifted from the current model
- ▶ so MoCo uses two encoders, θ and ϕ , one for z_i and one for z_k ($\forall k \neq i$)
 - ▶ the second encoder is updated slowly using an exponential moving average (EMA)

$$\phi \leftarrow \beta \phi + (1 - \beta) \theta \quad (\text{with } \beta \approx 0.999)$$

- ▶ this keeps embeddings in the queue consistent

Avoiding Collapse

- ▶ self-supervised learning always runs the risk of finding trivial solutions, e.g., all input examples collapse to the same representation ($\forall x \in \mathcal{X}, x \mapsto z_0$)
- ▶ contrastive learning attempts to avoid collapse through
 - ▶ feature normalisation
 - ▶ negative sampling[†]
 - ▶ student-teachers branches with EMA
 - ▶ strong data augmentation
 - ▶ projection head between the representation and loss
- ▶ some advanced techniques promote latent structure, e.g., orthogonality, in the latent subspace to stop collapse (e.g., Li et al., “CLOP”, 2024)
- ▶ forcing the model to be able to reconstruct the input (i.e., auto-encode) is another, albeit older, technique for preventing collapse

[†]Some contrastive methods, e.g., BYOL, don't use negatives.

applications

Word2Vec (Skip-grams)

(Mikolov et al., 2013)

- ▶ recall from Week 8 ...
- ▶ we want word w_i to be a good predictor of nearby words $w_j \in \mathcal{N}_i$, i.e., we want to maximize

$$\sum_{w_i \in \mathcal{T}} \sum_{w_j \in \mathcal{N}_i} \log P(w_j \mid w_i)$$

where P is modelled as the **softmax** over embedding similarity, i.e.,

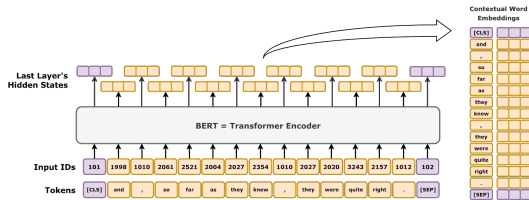
$$P(w_j \mid w_i) \propto \exp(z_i^T z_j)$$

- ▶ each word in the vocabulary $w_i \in \mathcal{V}$ is mapped to a learned latent representation $z_i \in \mathbb{R}^n$
 - ▶ equivalent to encoding w_i using a one-hot vector and passing through a linear layer, $z_i \leftarrow Aw_i + b$ where $A \in \mathbb{R}^{n \times |\mathcal{V}|}$
- ▶ in some cases z_i and z_j come from different encoders

BERT: Bidirectional Encoder Representations from Transformer

(Devlin et al., 2018)

- ▶ unlike Word2Vec, BERT encodes and entire sequence
 - ▶ i.e., token embeddings are *contextualized* — the same token can have a different embedding in different sentences
- ▶ uses an encoder-only transformer with “bi-directional” attention (no causal mask)
- ▶ trained by masked token prediction and next sentence prediction (binary)
- ▶ the <cls> token can be used as a sentence-level summary



- ▶ later methods (e.g., SimCSE, 2021) augment BERT with a contrastive loss to bring semantically similar sentences together and push different ones apart

CLIP: Contrastive Language-Image Pretraining

(Radford et al., 2021)

- embeds text T and images I in same latent space

$$z_i^{(I)}, z_j^{(T)} = f^{(I)}(I_i), f^{(T)}(T_j)$$

$$\ell = -\log \frac{\exp \text{sim}(z_i^{(I)}, z_i^{(T)})}{\sum_j \exp \text{sim}(z_i^{(I)}, z_j^{(T)})}$$

- contrastive variant SigLIP (2023)

$$\ell_{ij} = \begin{cases} -\log \sigma(\text{sim}(z_i^{(I)}, z_j^{(T)}) + b) & i = j \\ -\log \sigma(-\text{sim}(z_i^{(I)}, z_j^{(I)}) - b) & i \neq j \end{cases}$$

where σ is the logistic function

